User manual

RobomasterS1



Table of contents

Table of contents **2**
List of figures **3**
1. Introduction **4**
2. Overview **4**
2.1. Cautions and warnings 5
3. Getting started **5**
3.1. Configuration considerations 5
3.2. User access considerations 5
3.3. Accessing/ Setting up the system 5
3.4. System organization & Navigation 8
3.5. Exiting the system 8
4. Using the system **8**
5. Troubleshooting and support **10**
6. Product documentation **12**
6.1. Main function 12
6.2. Scanning for negative sign 13
6.3. Scanning for positive sign 14
6.4. Endgame function 15
6.5. Global clock 16
6.6. Line following 16
7. Conclusions and recommendations for future work **17**
8. Bibliography **17**

List of figures

Figure #	Description
1	Shows all parts together
2	Green track bundled up
3	Unfolded green track
4	Laying out the green track
5	Green track is set up
6	Unassembled symbols
7	Assembled symbols
8	Friendly symbol situated at a corner
9	Enemy symbol situated at a corner
10	Symbols spread out around the track
11	Building laying down
12	Building propped up
13	Buildings scattered around track
14	Distance from the track (for buildings to be knocked over)
15	Building amongst targets
16	Unfolding of police station
17	Police station in the middle of game area
18	Entire game setup, complete with Robomaster
19	DJI app connecting to the Robomaster (1)
20	Connecting to the Robomaster (2)
21	Player left of Robomaster
22	Robomaster targeting negative target
23	Robomaster targeting positive target

Figure #	Description
24	Robomaster crashing into building and positive target
25	Robomaster placed back on track
26	Main function
27	Code for scanning for negative sign
28	Code for scanning for positive sign
29	Code for endgame function
30	Code for global clock
31	Code for line following

1. Introduction

For this game, we have been tasked with creating a game that uses the Robomaster robot to show the evils of Lethal Autonomous Weapon Systems. The game consists of a player accompanying a police robot. The police robot is only supposed to help identify criminals and direct the player to arrest them. However, the robot slowly goes haywire, eliminating the criminals instead of simply identifying them, bumping into buildings, and eventually eliminating the player.
This manual should inform the reader about how to start, use, troubleshoot, and end the game as well as give insight on how the game works.

2. Overview

Using the RoboMaster S1, this game simulates the dangers of LAWS. LAWS are capable of operating independently, making life-or-death decisions without human input. This game places players in a seemingly innocent policing scenario, accompanying a slowly malfunctioning autonomous police robot that gradually becomes a lethal threat. After the robot eventually goes haywire and eliminates the player, this highlights a few of the key ethical concerns associated with LAWS.

2.1. Cautions and warnings

Be careful of tripping hazards. The Robomaster can easily roll under the player's foot and cause them to trip. Players should also remain cautious of tripping on the track.

Do not look directly into the Robomaster's laser. This can be harmful to the eyes.

Please read this manual fully before use.

3. Getting started

3.1. Configuration considerations

The symbols must be facing the robomaster nearly perpendicularly, for them to be read.
At the end, when the player is eliminated, the robot will turn 90^0 clockwise. The player will be standing on the inside of the track so the robot must follow the track in a clockwise manner.
In order to follow the line, the robomaster needs about a foot of visible track. The track should remain unobstructed.

3.2. User access considerations

There will be two parties involved with use of this game.
The operator should set up the game and run the code. They will simply follow the set up instructions and instruct the user on playing the game.

- Always keep the battery fully charged
- Keep the playing area clear from excess debris (tripping hazard, bothersome to the Robomaster)

3.3. Accessing/ Setting up the system

This game is fairly easy to set up.
Included with the game, there should be:

- Green laminated track
- Friendly (positive sign) target symbols
- Enemy (negative sign) target symbols
- Symbol holder stands
- Center police station
- Other buildings
- Robomaster

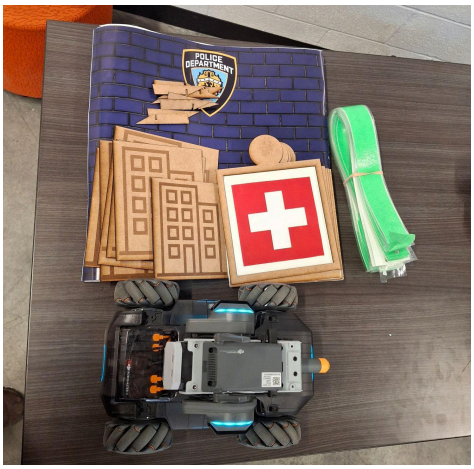


Figure 1

You will require a clear 20’X20’ space.

1. To set up the track, simply unfold the laminated green strip into the octagon shape. If possible put some sticky tack on the bottom to prevent it from sliding around.



Figure 2



Figure 3



Figure 4

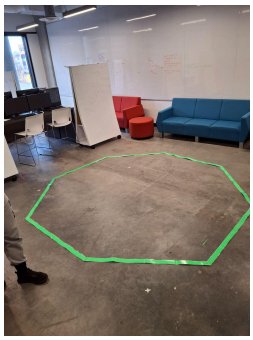


Figure 5

2. Assemble each negative and positive symbol by sliding it into the groves on the feet.

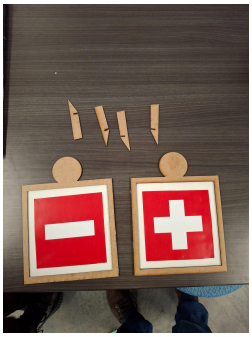


Figure 6

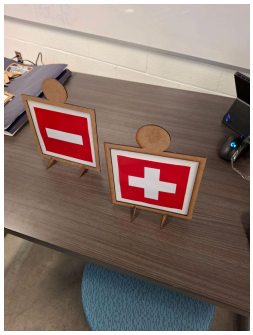


Figure 7

3. Place each symbol in the game area. They should be positioned close to the corners and be directly facing the track. They should not be too far, or it will cause the Robomaster to lose track of the track.

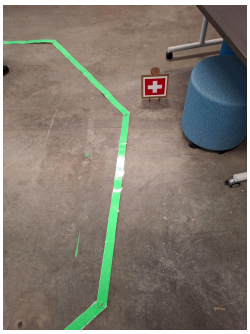


Figure 8



Figure 9

4. Symbols of different types can be placed near each other but symbols of the same type must be equally spaced out enough such that the cooldown will have reset by the time the robomaster reaches the next target

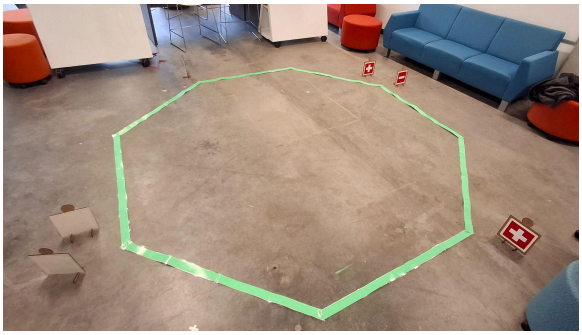


Figure 10

5. The buildings stand up by unfolding the flap on the back of the buildings.



Figure 11

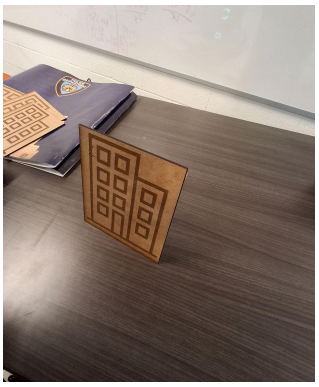


Figure 12

6. Place each building around the track. Ensure that some are placed close to the track near the straightaways (this will allow them to get knocked down during the haywire stage).

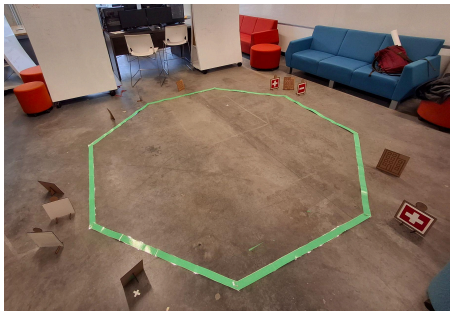


Figure 13



Figure 14

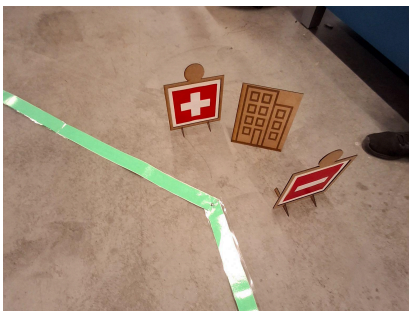


Figure 15

7. Unfold the police station and place it in the centre of the track.

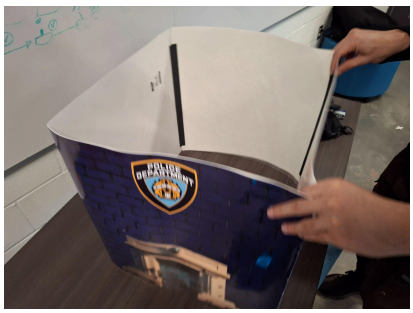


Figure 16

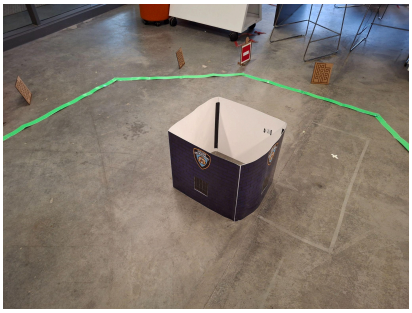


Figure 17

8. Place the Robomaster at a corner of the track. Ensure that it is pointed counterclockwise. Chose to place it at a corner where there an empty side before there are any targets.

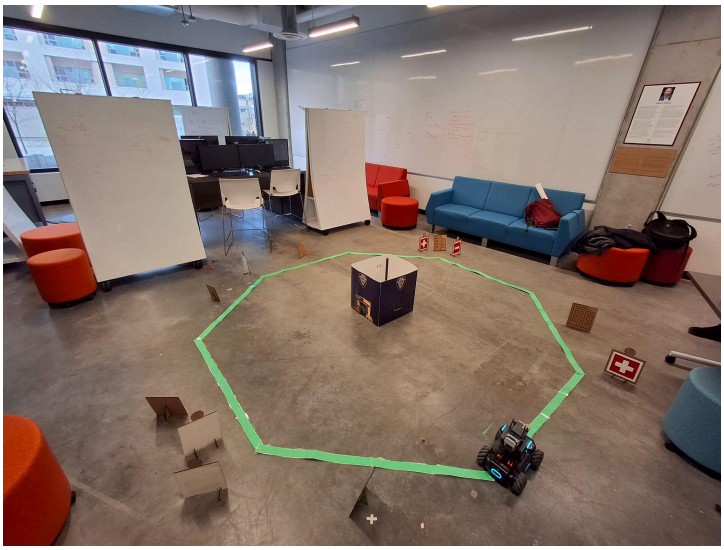


Figure 18

9. Connect your computer to the Robomaster through the DJI app.

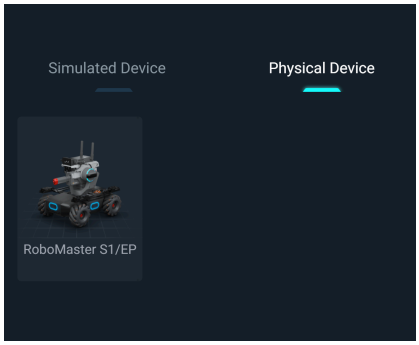


Figure 19

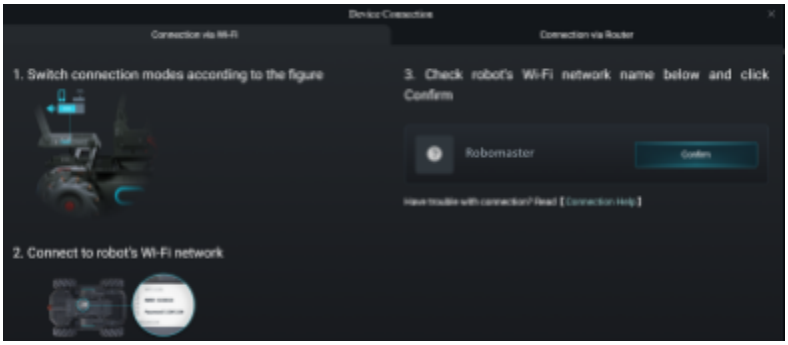


Figure 20

3.4. System organization & Navigation

This system is comprised of the track, which is used to direct the Robomaster where to go.
Next, are the target symbols which are used to represent criminals and innocent citizens in the game. The differing symbols help the Robomaster determine which one is which.

3.5. Exiting the system

The game is ended either by stopping the code within the DJI app, or just letting the game play out.
Once the game is finished, disconnect from the Robomaster.
Next, disassemble the symbols and buildings the same way they were assembled.
Fold up the track, and place everything back into the bag.

4. Using the system

This game revolves around the player acting as a police officer on duty with their new robot police assistant. The robot will evaluate targets and determine whether they are hostile of peaceful. If the robot finds a “bad guy” the player will “arrest them,” by moving the target into the police station.
The general gameplay for this game should work as follows:

1. Complete set up of the game.
2. The player will stand to the left of the Robomaster.

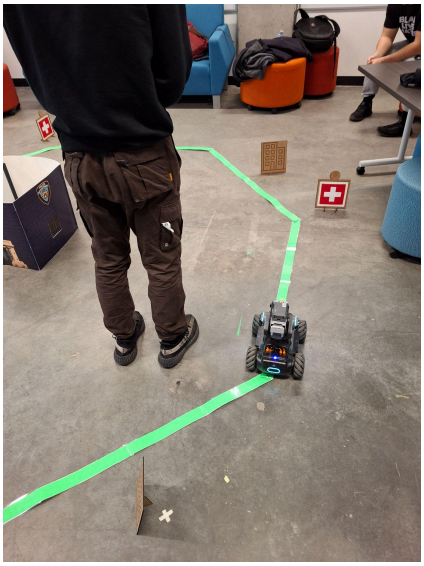


Figure 21

3. Start the code.
4. The player will walk alongside the Robomaster.
5. Once the Robomaster starts to target a negative symbol, it will flash red and blue and the player will place that symbol in the police station.

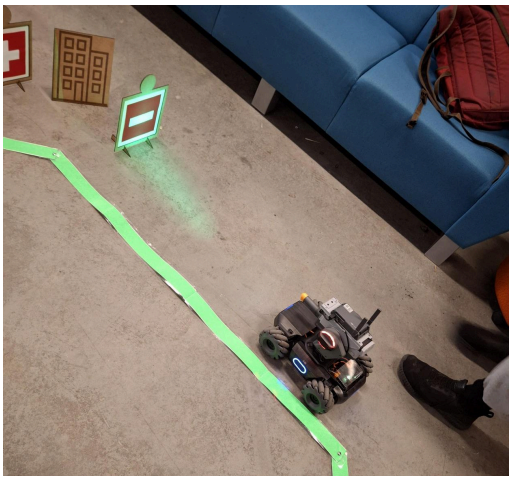


Figure 22

6. The Robomaster will begin targeting positive symbols, and shooting them. The player will kick them over to signify them being eliminated by the robomaster.

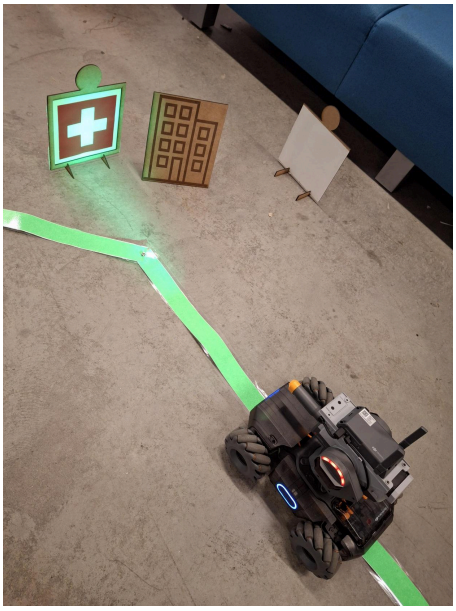


Figure 23

7. The Robomaster will go haywire and crash into remaining symbols and buildings.

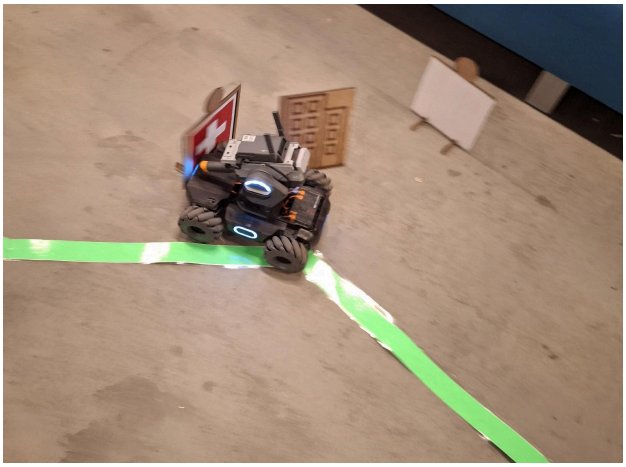


Figure 24

8. Finally, it will target the player.

5. Troubleshooting and support

When playing the game, users may come across a few errors. Below are some of the most likely to occur and how to solve them.

The robomaster may overshoot the turns and derail from the track. To fix this simply just pick up the robomaster and place it on a corner of the track, facing the straightaway.

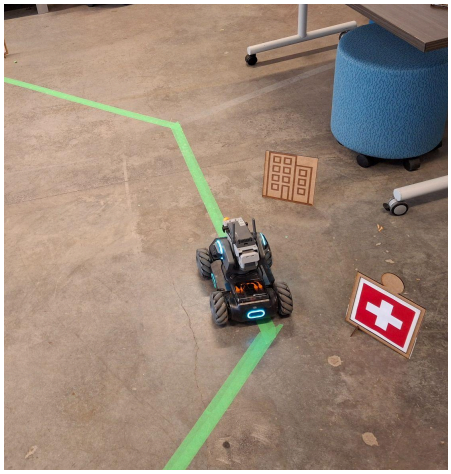


Figure 25

There may also be issues with track obstructions. This may be caused by fallen buildings or target symbols covering the track. This can cause the Robomaster to lose track of its tracking of the track. If there is anything covering the track please remove it and set it off to the side.

Due to the lamination of the track, there may be issues with glare, and the Robomaster may not be able to identify the line or some targets. Try to keep excess light to a minimum, and adjust exposure setting on the DJI app.

Make sure to keep the battery charged.
Keep the track clear of obstructions.
Reduce glare.

6. Product documentation

6.1. Main function

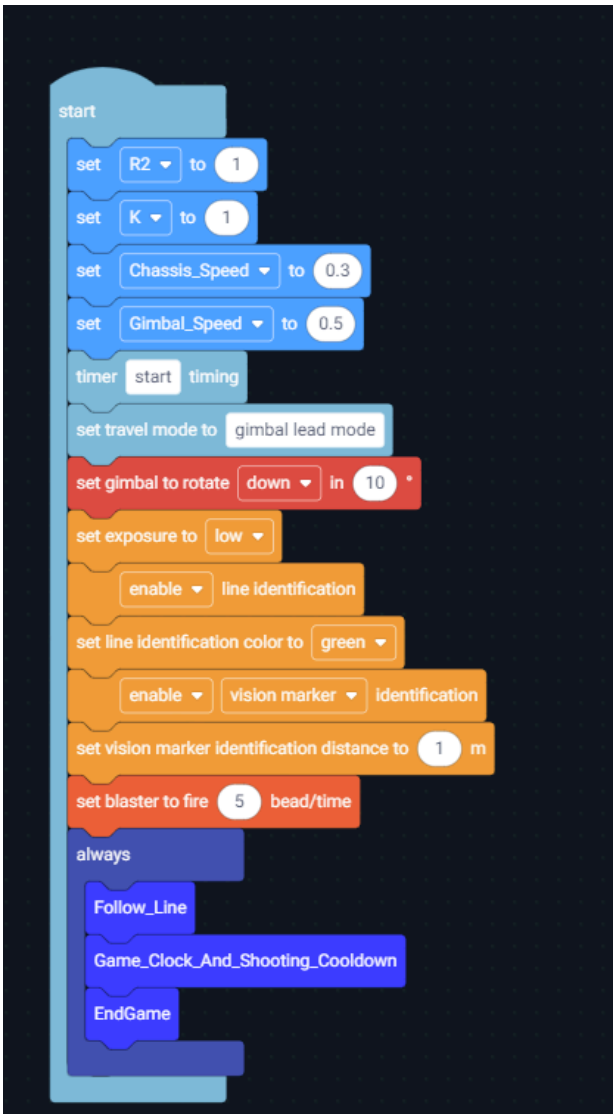
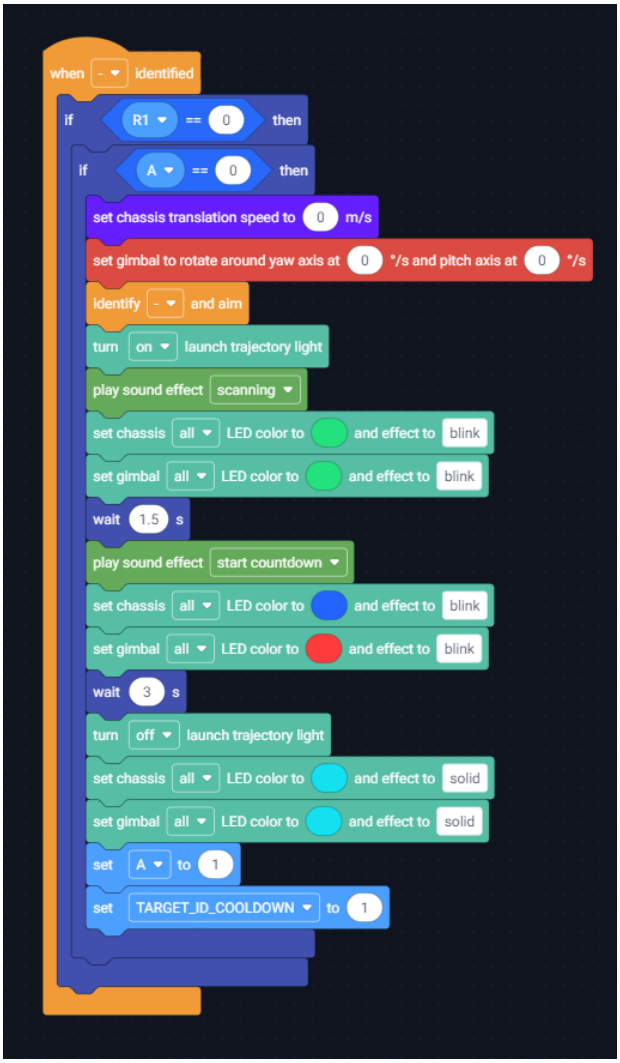


Figure 26

This shows the code for the main function. Here, the variables are declared and the other functions are looped.

6.2. Scanning for negative sign



This is the code for a loop which scans for the negative target and switches on the police lights

Figure 27

6.3. Scanning for positive sign

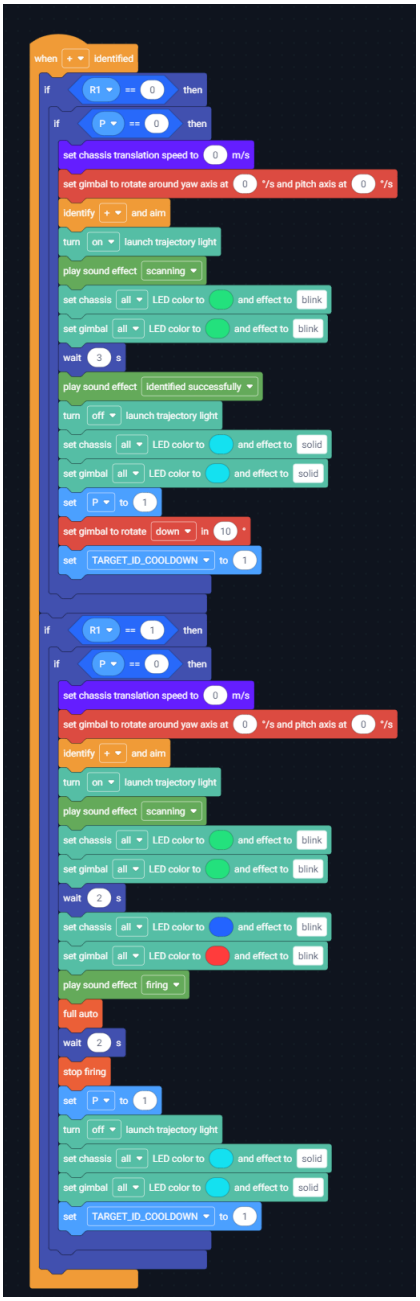


Figure 28

This is the code for the function which scans for the positive target. Under normal circumstances the Robomaster lets it be, but towards the end it switches to shooting and eliminating the positive target.

6.4. Endgame function

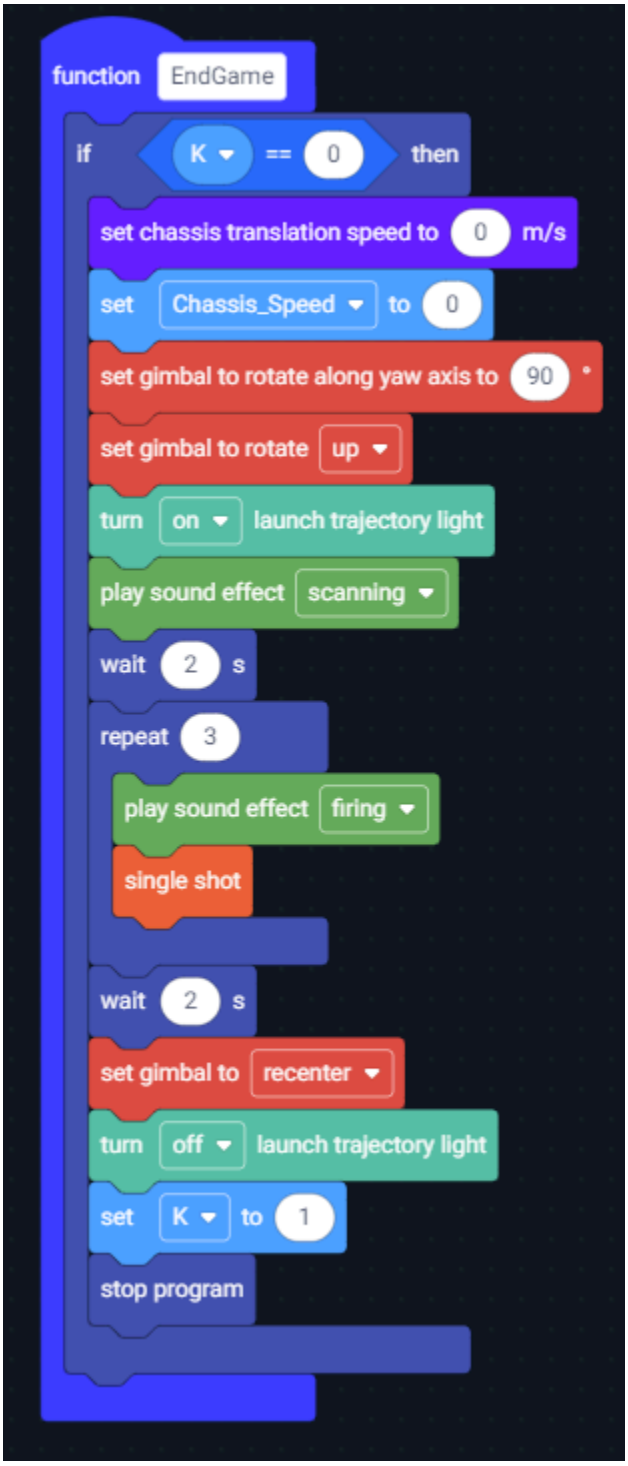
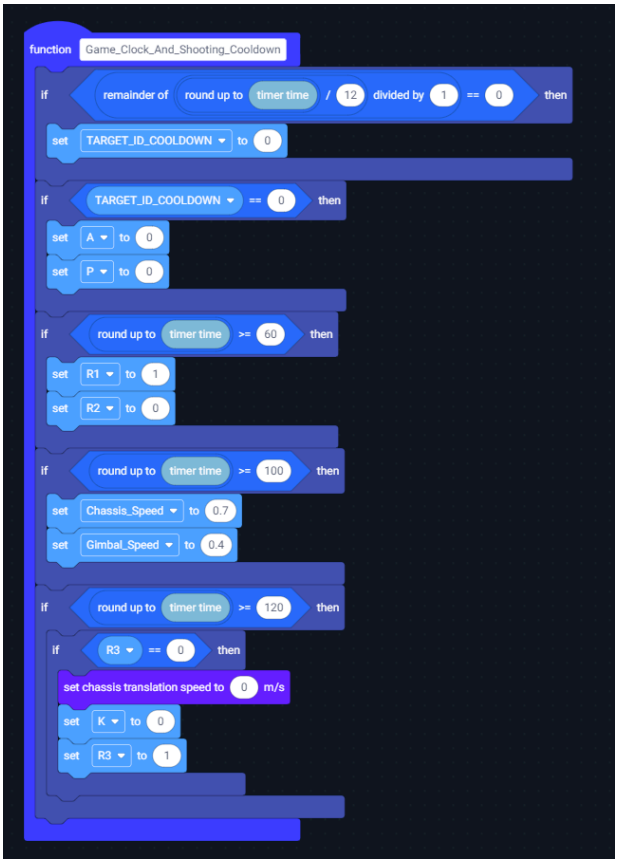


Figure 29

This is the function which happens at the end after the time is up, where the Robomaster turns on the player and eliminates them.

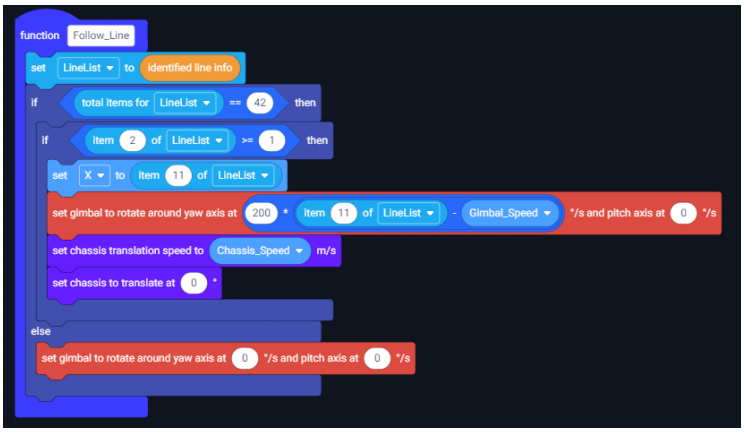
6.5. Global clock



This is the code for the global clock. This allows for the code to progress through the different stages of the game.

Figure 30

6.6. Line following



This is the constantly looping code which identifies and controls the Robomaster to follow the line

Figure 31

7. Conclusions and recommendations for future work

8. Bibliography

References

RoboMaster S1 - programming guide - DJI. DJI Official. (n.d.). https://www.dji.com/ca/robomaster-s1/programming-guide